

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2024–2025
FULL MARKS: 120

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Simplify the following expression using the cut-and-try method and show that: 8 + 4
(CO2)
(PO1)
 - i) $F_1(a, b, c, d) = a + \bar{a}b + \bar{a}\bar{b}c + \bar{a}\bar{b}\bar{c}d = a + b + c + d$
 - ii) $F_2(X, Y, Z) = XY + \bar{X}Z + YZ = XY + \bar{X}Z$
- b) Simplify the following Boolean expression using the QuineMcCluskey (tabular) method. 18
(CO3)
(PO2)

During the process, identify the prime implicants and essential prime implicants. Justify your choice of essential prime implicants in the final answer.

$$F(A, B, C, D) = \prod M(3, 4, 11, 12, 13, 15), d(7, 6)$$
2. a) A combinational circuit converts the 4-bit binary representation of a **valid decimal digit** into its 4-bit 2's-complement representation. 4 + 4 ×
4 + 5
(CO5)
(PO3)
 - i) Construct the complete truth table, including invalid BCD inputs and treat the outputs corresponding to invalid inputs as don't-care conditions. (The conversion of each binary input to its 4-bit two's complement representation need not be shown.)
 - ii) Using Karnaugh maps, obtain the minimized Boolean expression for each output bit.
 - iii) Draw the logic implementation of the simplified expression of each output variable using only **NAND gates**.
- b) Convert $(10010110)_{\text{GRAY}}$ into its corresponding binary representation and determine whether the resulting binary number has even parity. 5
(CO1)
(PO1)
3. a) Convert the following decimal numbers into their corresponding BCD (Binary-Coded Decimal) representations. Then perform BCD addition to find the sum of the numbers. 10
(CO2)
(PO1)

$$(2856)_{10}, (7639)_{10}, (4562)_{10}$$
- b) Perform the subtraction $(43307)_8 - (20431)_{10}$ by first converting both numbers into **base 12 number system**. Then carry out the subtraction using the **radix complement method** in **base 12**. You can skip the calculation of conversion of numbers between known bases and directly use values obtained from a calculator. 10
(CO1)
(PO1)
- c) The following calculation is performed in a number system with base x : 10
(CO1)
(PO1)

$$(15)_x + (4)_x = (22)_x$$

Find the value of the base x .
4. a) Draw a 3-bit Adder-Subtractor circuit using Full Adders (FA). You may represent the Full Adders as block diagrams and do not need to show their internal gate-level implementation. 4 + 6
(CO3)
(PO2)

Illustrate the operation of the circuit using two numerical examples, one demonstrating addition and the other demonstrating subtraction.

- b) You are required to design a combinational circuit that takes two 4-bit binary numbers A and B as inputs and produces the result of the subtraction $A - B$ using the **diminished radix complement method**. The following conditions must be considered while designing the circuit:

12 + 3
(CO5)
(PO3)

- B is always subtracted from A , but B is not necessarily smaller than A .
- You may use block diagrams of standard components such as a 4-bit adder (which accepts two 4-bit inputs, a carry-in, and produces a 4-bit sum and a carry-out), along with any required logic gates.
- The output must consist of a single 4-bit value representing the magnitude of the difference, along with one additional bit indicating the sign of the result (positive or negative). Separate outputs for positive and negative results are not allowed.

Also, add a short description of the logic behind the construction of the circuit.

- c) Is the circuit shown in Figure 1 a combinational or a sequential circuit? Justify your answer. Also, suggest appropriate modifications to convert it into the other type of circuit.

3 + 2
(CO3)
(PO2)

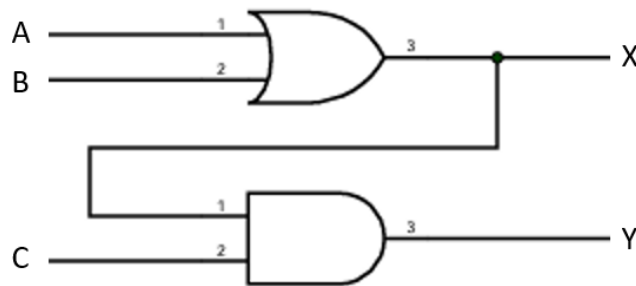


Figure 1: Circuit diagram for Question 4.c